Solution:
There are 2 roads connecting Chennai to Trichy. Suppose these are R_1 and R_2. Further there are 3 roads connecting Trichy to Tirunelveli . Let us name them as S_1, S_2 and S_3. Suppose a person chooses R_1 to travel from Chennai to Trichy and may further choose any of the 3 roads S_1, S_2 or S_3 to travel from Trichy to Tirunelveli. Thus the possible road choices are $(R_1, S_1), (R_1, S_2), (R_1, S_3)$. Similarly, if the person chooses R_2 to travel from Chennai to Trichy, the choices would be $(R_2, S_1), (R_2, S_2), (R_2, S_3)$.

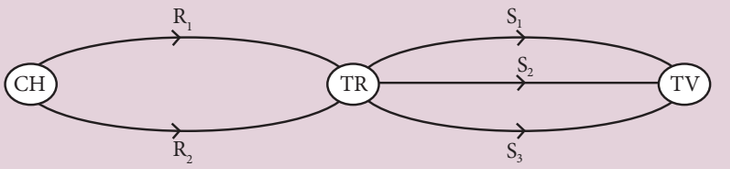


Figure 4.1

Thus there are $2 \times 3 = 6$ ways of travelling from Chennai to Tirunelveli.

An extension of the product rule may be stated as follows:
If a task comprises of n procedures $P_1, P_2, P_3, \cdots, P_n$ which can be performed in $m_1, m_2, \cdots, m_n$ ways respectively, and procedure P_i can be done after procedures $P_1, P_2, P_3, \cdots, P_{i-1}$ are done, then the number of ways of completing the task is $m_1 \times m_2 \times \cdots \times m_n$.

3. **The Inclusion-Exclusion Principle** Suppose two tasks A and B can be performed simultaneously. Let $n(A)$ and $n(B)$ represent the number of ways of performing the tasks A and B independent of each other. Also let $n(A \cap B)$ be the number of ways of performing the two tasks simultaneously. We cannot use the sum rule to count the number of ways of performing one of the tasks as that would lead to over counting. To obtain the correct number of ways we add the number of ways of performing each of the two tasks and then subtract the number of ways of doing both tasks simultaneously. This method is referred to as the principle of inclusion - exclusion. Using the notation of set theory we write it as

$$n(A \cup B) = n(A) + n(B) - n(A \cap B).$$

Suppose we have to find the number of positive integers divisible by 2 or 7 (but not both), upto 1000. Let $n(A)$ denote the number of integers divisible by 2, $n(B)$ denote the number of integers divisible by 7 and $n(A \cap B)$ the number of integers divisible by both 2 and 7. Then the number of positive integers divisible by 2 or 7 is given by

$$n(A \cup B) = n(A) + n(B) - n(A \cap B) = 500 + 142 - 71 = 571.$$

(Note that $n(A)$ will include all multiples of 2 upto 1000, $n(B)$ will include all multiples of 7 upto 1000 and so on.)

Tree Diagrams: Tree diagrams are often helpful in representing the possibilities in a counting problem. Typically in a tree the branches represent the various possibilities. For example, suppose a person wants to buy a Car for the family. There are two different branded cars and five colours are available for each brand. Each colour will have three different variant on it namely GL,SS,SL. Then the various choices for choosing a car can be represented through a tree diagram as follows: